

Summary of Key Findings: Training a Transformer-Based Language Model

1. Project Overview & Baseline Performance The objective of this research was to train a GPT-style decoder-only transformer language model from scratch and analyze the impact of various architectural and hyperparameter decisions. The model was trained on *The Adventures of Sherlock Holmes* from Project Gutenberg using a 128 embedding dimension, 3 transformer layers, 4 attention heads, and a context length of 32 tokens.

- **Findings:** The baseline model successfully learned token-level statistical patterns, generating domain-relevant text. However, validation loss eventually diverged from training loss, indicating expected overfitting due to the model's capacity relative to the small dataset (~100K tokens).

2. Experiment 1: Varying Epochs and Learning Rate The baseline architecture was tested across different learning rates (5e-5 to 1e-3) and epoch counts (5 vs. 10) to evaluate convergence stability.

- **Findings:** A learning rate of 1e-4 over 5 epochs provided the best balance between stable optimization and generalization. Higher learning rates (5e-4, 1e-3) caused unstable training and frequent minima overshooting. Increasing epochs to 10 at the optimal learning rate simply accelerated overfitting, causing the model to memorize the training data rather than generalize.

3. Experiment 2: Varying Transformer Layers To study the effect of architectural depth, models were trained with 1, 3, 5, 7, and 12 transformer layers.

- **Findings:** Shallow models (1 layer) underfit the data due to a lack of capacity to capture complex contextual dependencies. Conversely, excessively deep architectures (7 and 12 layers) suffered from severe overfitting, memorizing the limited dataset quickly and yielding the worst validation performance. Configurations with 3 to 5 layers offered the optimal tradeoff between representation learning and generalization.

4. Experiment 3: Varying Attention Heads The multi-head attention mechanism was tested using 1, 2, 4, and 8 heads, keeping the embedding dimension fixed at 128.

- **Findings:** A single attention head resulted in slow convergence and poor validation loss, as it could only capture limited contextual relationships. The 4-head configuration performed best. Scaling up to 8 heads produced diminishing returns because dividing the fixed 128-dimensional embedding across too many heads reduced the representational capacity available to each individual head.

5. Experiment 4: Ablation Studies Core components were selectively removed to assess their contribution to the network.

- **No Normalization Layers (High Severity):** Caused inconsistent gradients and highly unstable optimization, resulting in poor generalization and repetitive inference text.
- **No Residual Connections (High Severity):** Severely degraded convergence stability. The lack of shortcut pathways obstructed gradient flow, resulting in higher losses and contextually incoherent text generation.
- **No Feedforward Neural Network (Medium Severity):** While the model could still learn via the attention mechanisms, removing the FFN eliminated critical nonlinear feature transformations, reducing the network's overall representational capacity and raising the baseline loss.